

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I G. Srivalli (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G. Srivalli

Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. **Scenario: Event Communication Security:** A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. **Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.**
- b. **Explain how an authorized member, aware of the encoding rule, would retrieve the original message.**

Solution:

Event Communication Security – Caesar Cipher:

Given: Plaintext message = "MEET AT NOON", Shift (key) = 4

(a) Encrypting the Message:

This encryption method is the Caesar cipher, a substitution cipher in which every letter of the message is shifted forward by a fixed number of positions in the alphabet (here, 4 positions). Spaces remain unchanged.

Encryption rule: $C = (P + 4) \bmod 26$, where P is the position of the plaintext letter (A = 0, B = 1, ..., Z = 25) and C is the position of the ciphertext letter.

Letter-by-letter encryption: Table:

Plaintext	M	E	E	T		A	T		N	O	O	N
+4 shift	Q	I	I	X		E	X		R	S	S	R

Encrypted message (ciphertext): **QIIX EX RSSR**

(b) How an Authorized Member Decrypts It:

Since the authorized member knows the encoding rule (a fixed shift of 4), decryption is simply the reverse operation: shift every letter backward by 4 positions in the alphabet, i.e. $P = (C - 4) \bmod 26$.

Working through it:

- Q → M, I → E, I → E, X → T → "MEET"
- E → A, X → T → "AT"
- R → N, S → O, S → O, R → N → "NOON"

So, the member recovers the original message: "MEET AT NOON".

This works because the Caesar cipher is symmetric — the same shared secret (the shift value of 4) is used for both encryption and decryption. Anyone without knowledge of the key sees only the meaningless string "QIIX EX RSSR," while authorized members who know the shift can instantly reverse it.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: **"WECRLTEERDSOEFEAOCAIVDEN"**.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Solution:

Legacy Encryption in Corporate Communication – Rail Fence (Transposition) Cipher

Given: Ciphertext = "WECRLTEERDSOEFEAOCAIVDEN" (25 letters),

three levels of arrangement → 3-rail fence cipher

(a) Reversing the Encryption Process:

"Three levels of arrangement" indicates the classic rail fence cipher with 3 rails. In encryption, the plaintext is written in a zigzag across 3 rows (rails) and then read off row by row. To decrypt, we reverse this.

Step 1 – Determine rail lengths:

With 25 letters on 3 rails, the zigzag pattern (1,2,3,2,1,2,3,2,...) distributes the letters as:

Rail 1 = 7 letters, Rail 2 = 12 letters, Rail 3 = 6 letters.

Step 2 – Split the ciphertext into rails:

Table:

Rail	Letters
Rail 1	W E C R L T E
Rail 2	E R D S O E E F E A O C
Rail 3	A I V D E N

Step 3 – Place the letters back into the zigzag grid:

W				E				C				R				L				T				E
	E		R		D		S		O		E		E		F		E		A		O		C	
		A				I				V				D				E				N		

Step 4 – Read along the zigzag path: (top → middle → bottom → middle → top ...):

W-E-A-R-E-D-I-S-C-O-V-E-R-E-D-F-L-E-E-A-T-O-N-C-E

Original message: **"WE ARE DISCOVERED FLEE AT ONCE"**

(b) Why This Method Is Vulnerable in Modern Communication Systems:

- **Tiny key space:** The only secret is the number of rails, so an attacker can simply try every possibility (2, 3, 4, ...) — even by hand.
- **Letters are only rearranged, not changed:** The ciphertext has the same letter frequencies as the plaintext, making frequency analysis and anagramming easy.
- **Security by obscurity:** Once the method is known, everything is exposed — violating Kerckhoffs's principle, which says security must depend on the key, not on hiding the algorithm.
- **No integrity protection:** An interceptor can alter or rearrange the message without the receiver noticing.
- **No match for modern computing:** A hand cipher from the pre-computer era offers essentially zero protection today.

(c) Proposed Enhancement:

Replace the transposition cipher with AES-256-GCM (a modern authenticated encryption standard). AES-256 applies multiple rounds of substitution and permutation with a 256-bit key, making brute-force attacks computationally infeasible, while GCM mode adds an authentication tag so tampering is detected. It is fast, standardised, and available in all major software libraries, so it can be integrated into the legacy system with a secure key-management process.